

Gieni Platform Reference Architecture (v2.0)

Status: Approved

Category: System Architecture

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Classification: Canonical Engineering Architecture | Version: 2.0 | Status: Production Approved

1. The Nine Autonomous Production Microservices

- **Property Service:** Reconciles petitions with Assessor rolls & PostGIS. Emits Property.Identified.
- **Ownership Service:** Audits deed instruments, vesting, and net equity waterfall. Emits Ownership.Updated.
- **Control Service:** Maps multi-heir social networks, de-facto decision maker, and Neo4j graph edges. Emits Control.Changed.
- **Authority Service:** Tracks docket minutes and classifies signatory capacity (Tiers 1-4). Emits Authority.Updated.
- **Opportunity Service:** Computes Deal Friction Score (DFS) and viability score (0-100). Emits Opportunity.Scored & QC events.
- **Evidence Service:** Maintains chain-of-custody PDF storage and SHA-256 invariant hash commit. Emits Evidence.Ingested.
- **Client Service:** Governs single-partner county exclusivity, capacity quotas, and buy-box matching.
- **Delivery Service:** CRM webhook dispatcher with 5 retries, DLQ fallback, and Twilio SMS Priority A alerts.
- **Workflow & Orchestration Service:** Temporal/BullMQ DAG lifecycle state machine manager and exception routing engine.

2. 14-Stage Opportunity Lifecycle Progression

Part 1: Research Stages: DISCOVERED -> PROPERTY_IDENTIFIED -> OWNERSHIP_RESOLVED -> CONTROL_MAPPED -> AUTHORITY_RESOLVED -> SCORED -> QC_CERTIFIED.

Part 2: Commercial Stages: DELIVERED -> CONTACTED -> APPOINTMENT -> OFFER_PRESENTED -> CONTRACT_EXECUTED -> CLOSED_WON / CLOSED_LOST -> ARCHIVED.

3. Deterministic Failure Recovery Runbooks

1. Municipal Scraper: 3 retries (1m, 5m, 15m) + proxy rotation -> PagerDuty escalation.
2. OCR Parsing: Corruption diversion to Tasks & Exceptions DB.
3. Title Conflict: Broken deed chain isolation.
4. Authority Ambiguity: Caveat/conflict legal queue routing.
5. Partner Webhook: 5 retries over 2 hours -> SMS/Email fallback -> Dead-Letter Queue (DLQ).

